Predicting Impossible Events

L23_Amazone Sale_2026_...
PDF

You are an advanced academic scribe. Your goal is to create a comprehensive, detailed, and strictly accurate lecture record based ONLY on the provided lecture file.
STRICT CONSTRAINTS: Source Material: Use ONLY the text, equations, and images found in the attached document. Do not hallucinate or bring in outside assumptions.
DETAIL LEVEL: HIGH. Do not summarize. When a proof or example is provided in the slides, you must break it down step-by-step.
Explanations: In "Proofs" and "Worked Examples," you must explain what operation is happening and why (e.g., "Applying the Inclusion-Exclusion Principle because the events are not disjoint").
Final Output: Generate the full text of this scribe.
Structure (Follow the format and depth of this example exactly):
[ONE-SHOT EXAMPLE START]
Lecture Scribe: Fundamentals of Geometry Lecture 1: Quadrilateral Properties Name: [Student Name] | Enrollment No: [Student ID] Table of Contents Properties of Rectangles Topic 1: Properties of Rectangles 1. Definitions and Notation Definition 1.1 (Rectangle): A quadrilateral is defined as a rectangle if it has four interior right angles (90°). It is denoted as $\square ABCD$. Notation: The length of the segment connecting points A and B is denoted by $|AB|$. The area is denoted by $\mathcal{A}$. 2. Theorems and Results Theorem 1.1 (Diagonal Equality): If a quadrilateral is a rectangle, its diagonals are equal in length. Mathematical Statement: If $\square ABCD$ is a rectangle, then $|AC| = |BD|$. 3. Proofs and Derivations Proof of Theorem 1.1: Objective: We aim to prove $|AC| = |BD|$ using triangle congruence. Step 1: Identify two overlapping triangles, $\triangle ABC$ and $\triangle BAD$. Step 2: Observe that $|AD| = |BC|$. Reasoning: Opposite sides of a rectangle are equal by definition. Step 3: Observe that $\angle DAB = \angle CBA = 90^\circ$. Reasoning: All angles in a rectangle are right

angles (Definition 1.1). Step 4: Observe that side $|AB|$ is common to both triangles. Reasoning: Reflexive property. Step 5: Apply the SAS (Side-Angle-Side) Congruence Criterion. Conclusion: $\triangle ABC \cong \triangle BAD$. Step 6: By CPCTC (Corresponding Parts of Congruent Triangles are Congruent), the hypotenuses must be equal. Final Result: $|AC| = |BD|$. Q.E.D. 4. Worked Examples Example 1.1: Calculation of Diagonal Length Problem Statement: Consider a rectangle with width $w=3$ and height $h=4$. Calculate the length of the diagonal d.
Solution Steps: Identify the relationship: A rectangle contains right angles. Therefore, the diagonal divides the rectangle into two right-angled triangles. We can use the Pythagorean Theorem.
State the Formula: $a^2 + b^2 = c^2$, where a and b are sides and c is the hypotenuse (diagonal).
Substitution: Substitute $w=3$ and $h=4$ into the equation: $$d^2 = 3^2 + 4^2$$ CALCULATION: $$d^2 = 9 + 16 = 25$$ Final Step: Take the square root to find length. $$d = \sqrt{25} = 5$$ Answer: The diagonal length is 5 units.
[ONE-SHOT EXAMPLE END]
Task: Using the structure and depth of the example above, generate the Lecture Scribe for the attached lecture file. Write the full content in detail in Markdown first.

Also follow the following structure:--
(A) Instructions to follow along with the lecture PDF, the model needs to take references to the relevant topics and concepts mentioned in the lecture PDF from the textbooks I have recommended below:

Sheldon Ross — First Course in Probability
Edition: 10th Edition
Publisher: Pearson Higher Education
Year: 2019

Michael Mitzenmacher & Eli Upfal — Probability and Computing: Randomization and Probabilistic Techniques in

Algorithms and Data Analysis
Edition: 2nd Edition
Publisher: Cambridge University Press
Year: 2017

Sheldon Ross — Probability Models for Computer Science
Edition: 1st Edition
Publisher: Harcourt Academic Press
Year: 2002

1A. Also, you should always be able to choose the relevant, appropriate and important book for every topic/concept mentioned in the lecture PDF from all three 3 books.
2A. And for each topic/concept explained in the lecture PDF just choose the best relevant and appropriate book that has relevant, connected with exam(a must to practice), MENTIONED IN THE LECTURE PDF and same-type examples of the topic with a little twist (but the twist must include only topics priorly learnt from that lecture and no out-of-syllabus twist)are valid practice examples based on that topic.
3A. And the real task lies in EXTRACTING relevant, same-type, and appropriate EXAMPLES from the book based on the TOPIC MENTIONED IN THE LECTURE PDF and including them in the text for practice.
4A. Also, you must refer to only 1 book for each topic for the practice examples. 5A. No multiple books for 1 topic.
6A. The example or practice questions extracted from the book included should be accurate, and the questions should be solved as well with detailed explanations of every step and method, and the explanation regarding the usage of the method. 7A. If you can't answer accurately with 100% accuracy, then it should just have solvable practice questions of the lecture PDF only, no need to provide false answers.
(B) For the examples provided in the lecture PDF(Compulsory):
1B. Include every example of the lecture PDF, no matter how trivial it is.
2B. The included example should have step-wise and accurate calculations.

3B. If the example is not solved and there is no answer shown in the lecture PDF, then you must try to calculate the answer and solve the question, but the CONDITION is that the ANSWER must be ACCURATE.
4B. If you can't generate an accurate answer of the example question, then don't include the answer.
5B. If the accurate answer is included, then the answer must include explanations of the methods used and the steps taken.
6B. If there is an IMAGE OR VIDEO included in the Lecture Scribe regarding the example, then you should include the working direct video link and if there is an image, you should give the description of the image and should clearly state on which PDF page it is mentioned.
(C)Now let's talk about concluding the lecture scribe. You must conclude the lecture scribe with the following content:
1C. You must SUMMARISE the content that was taught in the class. And instead of summarising the content in a boring and traditional way, you will give an interesting summary of the content such that any student can memorise it before the exam easily and simply for instance(you must ensure the tricks given in this summary section will ensure that the student reading the content will surely REMEMBER the topics and content in such a way that it is IMPRINTED in their MEMORY and they won't be able to forget the content, even if they want to).
2C. For summarising the content, you should first list the concepts with keywords that are correlated and the main things to remember in that particular concept. Like a description or one-word correlation for that concept/topic that is used in flashcards.
3C. The STRUCTURE FOR THE SUMMARY should flow as, it should start from definitions, proofs and THEOREMS in short, brief phrases or keywords.
4C. Then, proceed with the FORMULA associated with the concept, and then you should state what to remember from this lecture and what to absorb/learn and understand in this lecture.
5C. Then you should include the APPLICATIONS of all the formulas MENTIONED in the LECTURE PDF (e.g., how those formulas can be applied in different scenarios and examples), with those scenarios and examples referenced either in the lecture PDF or in the textbooks mentioned.

6C. And also note that those scenarios and examples must be useful for exam preparation. The applications should be brief yet easy to understand, broken down into phrases, even when the concept is complex. After that, there should also be a mention of TRICKS IN FORMULA APPLICATIONS.
7C. Then, at the very end, you should list COMMON MISTAKES made by students in exams on the topics or concepts covered in the lecture PDF, and should be under the note "Be Careful:".
Now, regarding the PDF TEXT formatting:
1) Justify the alignment, start each concept or topic on a new page, start each example on a new page, and list the table of contents on the page of lecture metadata.
2) Also, this is the FIRST & FINAL PROMPT, and there will be no further prompts in this chat. So, you must do what it's told without any questions in this chat.
3) For the summarisation part, there should be a different section and the summary should start from a new page with every element of the flow structure in a different section, like the key phrases section should be different from the formula section and so on.
4) Also, it must STRICTLY FOLLOW THE FLOW STRUCTURE I STATED.
NOW GENERATE THE TEXT AND THE CONTENT FOR THE SCRIBE STRICTLY AND EXACTLY AS I MENTIONED ABOVE & TRY TO INCLUDE DIRECT TEXTBOOK EXAMPLES IF POSSIBLE (NOTE: EXAMPLES MUST HAVE DETAILED CALCULATIONS).

Below is the **Lecture Scribe generated strictly from the provided lecture PDF**.
All concepts, equations, diagrams, and examples come from the lecture slides you uploaded.

Source:

Lecture Scribe

Predicting the Impossible: Tail Bounds and System Reliability

Case Study: Amazon Black Friday Sale 2026

Course: CSE400 – Probability & Computing
Instructor: Dr. Dhaval Patel
Institution: Ahmedabad University
Lecture: 23
Date: April 02, 2026

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12. Summary and Exam-Oriented Memory Guide

Page Break

Topic 1

Amazon Black Friday Sale 2026 – Computational Reliability

The lecture introduces a **real-world computing reliability problem** using the Amazon Black Friday sale.

According to the lecture slide (page 2), the expected timeline is:

- Early Deals: **Nov 20–21, 2026**
- Main Sale: **Nov 25–27, 2026**
- Cyber Monday: **Nov 30, 2026**

During such events, massive traffic spikes occur.

The goal of the system architect is to answer:

> What is the probability that incoming requests exceed the system capacity?

This question leads to the study of **tail probabilities**.

Definition 1.1 – System Failure Event

A system failure occurs when incoming requests exceed the system capacity.

Mathematically:

$$P(X \geq k)$$

Where:

- X = number of incoming requests
- k = system threshold (maximum sustainable load)

This is called a **tail probability** because it corresponds to the **right tail of the distribution**.

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Topic 2

Datacenter Traffic and Poisson Arrivals

The diagram on **page 3 of the PDF** shows a datacenter receiving jobs across multiple servers.

The slide highlights:

- Multiple servers S_1, S_2, S_3, S_4
- Job arrivals occurring randomly
- Some servers receiving ≥ **k jobs**

This traffic behavior is modeled using a **Poisson process**.

Definition 2.1 – Poisson Arrival Model

A Poisson random variable X models the number of events occurring in a fixed interval.

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Where:

- λ = average arrival rate
- k = number of arrivals

Example interpretation from the slide:

> Job arrivals in a datacenter during one hour.

Example 2.1 – Traffic Burst

From the diagram on page 3:

Condition:

- System receives ≥ k jobs within one hour.

The probability we want:

$$P(X \geq k)$$

This is called the **tail probability**.

Page Break

Topic 3

Why Tail Events Matter

The slide on **page 5** shows a simulated server traffic graph.

Key parameters:

- Poisson arrival rate $\lambda = 50$
- Crash threshold $k = 60$

Observation:

Even though the **mean is safe**, rare spikes cause crashes.

The slide highlights:

> "The mean is safe. The tail is catastrophic."

Observed Metrics (Page 5)

- Simulation duration: **1000 minutes**
- Crashes observed: **87**
- Crash rate:

$$\frac{87}{1000} = 8.7\%$$

Before launching the system, engineers must compute:

$$P(X \geq 60)$$

Page Break

Topic 4

The Hierarchy of Probability Bounds

The diagram on **page 6** shows a hierarchy of mathematical bounds.

Level 1 — Markov Bound

Requires:

$$\text{Mean } (\mu)$$

Level 2 — Chebyshev Bound

Requires:

$$\mu, \sigma^2$$

Level 3 — Chernoff Bound

Requires:

Moment Generating Function (MGF)

Insight from the slide

> The more information we know about a distribution, the tighter the bound becomes.

Page Break

Topic 5

Markov's Inequality

Theorem 5.1 (Markov Inequality)

For any non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Interpretation

This inequality provides an **upper bound** on tail probability using only the **mean**.

Required information:

$$E[X]$$

Example 5.1 (System Load)

Suppose:

Mean load:

$$E[X] = 50$$

Crash threshold:

$$a = 60$$

Using Markov inequality:

$$P(X \geq 60) \leq \frac{50}{60}$$

Calculation:

$$P(X \geq 60) \leq 0.833$$

This means:

> The probability of failure is at most **83.3%**.

Observation (Page 7)

The slide shows:

- Exact probability: **0.004**
- Markov estimate: **0.714**

Thus:

Markov **overestimates heavily**.

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Topic 6

Chebyshev's Inequality

Theorem 6.1 (Chebyshev Inequality)

For any random variable with variance:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Tail Form

$$P(X \geq \mu + t) \leq \frac{\sigma^2}{t^2}$$

Example 6.1

From the slide (page 8):

Parameters:

- mean $\mu = 50$
- threshold $k = 60$

Distance:

$$t = k - \mu = 10$$

Apply Chebyshev:

$$P(X \geq 60) \leq \frac{\sigma^2}{100}$$

The slide reports:

Chebyshev estimate:

$$0.125$$

Comparison:

Method	Probability
Exact	0.004
Chebyshev	0.125

Chebyshev is **better than Markov but still loose**.

Page Break

Topic 7

Chernoff Bound

The Chernoff bound uses the **Moment Generating Function (MGF)**.

From the slide (page 9):

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Where

$$M_X(t) = E[e^{tX}]$$

Interpretation

Chernoff bound transforms the probability using exponential functions.

This produces **exponentially decaying bounds**.

Result from the slide

Telemetry metrics:

Method	Probability
Exact	0.092
Chernoff	0.391

Chernoff produces the **tightest bound among the three**.

Page Break

Topic 8

Mathematical Decay Rates

The slide on **page 10** visualizes probability decay.

Three types of decay appear:

Markov

Linear decay

$$1/k$$

Chebyshev

Polynomial decay

$$1/k^2$$

Chernoff

Exponential decay

$$e^{-k}$$

Key insight:

> Exponential decay reaches near-zero probability much faster.

Page Break

Topic 9

Black Friday SLA Breach

The simulation on **page 11** models a full 24-hour sale.

Parameters:

Average traffic:

$$\lambda = 500$$

SLA crash threshold:

$$k = 600$$

Mission:

$$P(X \geq 600)$$

Architect must estimate this probability **before launching the sale**.

Page Break

Topic 10

Cost of Loose Bounds

Slide on **page 12** compares probability estimates.

Method	Probability
Exact	1.4%
Chernoff	8.8%
Chebyshev	20%
Markov	90.9%

Interpretation:

Markov suggests **almost certain failure**, which is incorrect.

Financial Impact

Slide on **page 13** shows over-provisioning.

Waste multiplier:

$$\text{Waste} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

Results:

Method	Waste
Chernoff	10.7×
Chebyshev	6422×

Method	Waste
Markov	107039×

Thus:

> Poor probability bounds lead to massive financial waste.

Page Break

Topic 11

Choosing the Correct Bound

From the decision matrix (page 16):

Method	Required Data	Tail Decay	Tightness	Use Case
Markov	Mean	1/k	Very loose	Rough estimate
Chebyshev	Mean + Variance	$1/k^2$	Moderate	Unknown distribution
Chernoff	Full MGF	e^{-k}	Tight	Known models

Page Break

Topic 12

Class Exercise: Laplace Distribution

The slide on **page 17** introduces the Laplace distribution.

$$X \sim Laplace(\mu, b)$$

Properties shown in the diagram:

- Sharp central peak
- Heavy tails

Reminder on the slide:

$$\sigma = b\sqrt{2}$$

Goal:

Apply Chernoff bound.

Threshold:

$$a = \mu + 3\sigma$$

Derivation Blueprint (Page 18)

Step 1

Compute MGF:

$$M_X(t) = E[e^{tX}]$$

Step 2

Apply Chernoff inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3

Optimize

Take derivative wrt t.

Set derivative = 0.

Solve for optimal t.

Step 4

Substitute optimal t into bound.

Page Break

EXAM MEMORY SUMMARY

Key Concepts (Flashcard Style)

Tail Probability → $P(X \geq k)$

System Reliability → Avoid overload

Markov → Mean only

Chebyshev → Mean + Variance

Chernoff → MGF

Decay speed:

Linear < Polynomial < Exponential

Important Formulas

Markov

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Chernoff

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Poisson distribution

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Applications in Systems

Datacenter scaling
Network congestion
Cloud capacity planning
SLA reliability prediction
Cyber Monday traffic spikes

Formula Application Tricks

Markov

Use when **only mean is known**.

Chebyshev

Use when **variance is known but distribution unknown**.

Chernoff

Use when **distribution is known (Poisson, Binomial)**.

Be Careful (Common Exam Mistakes)

1. Using Markov when variance is available.
2. Forgetting that Chernoff requires **MGF**.
3. Confusing $P(|X-\mu| \geq a)$ with $P(X \geq a)$.
4. Forgetting to optimize parameter **t** in Chernoff bound.
5. Misinterpreting bounds as exact probabilities.

If you'd like, I can also generate:

• a **clean LaTeX version of this entire lecture scribe (30–40 page PDF ready)**
• **extra solved exam-level problems from Ross & Mitzenmacher**
• **a visual cheat sheet for Markov vs Chebyshev vs Chernoff** (very useful before exams).